

TUZLA, Bosnia and Herzegovina

WMO: 145570

Lat: 44.5420N

Lon: 18.6851E

Elev: 991

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	15.8	6.8	8.1	11.0	12.5	10.8	19.4	13.6	36.8	11.8	34.9	2.5	260	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	92.9	70.5	89.5	70.1	86.2	69.4	75.1	86.0	72.9	84.0	71.0	81.5	5.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	121.3	80.2	69.7	113.2	78.5	67.9	106.3	75.9	39.1	86.3	37.2	84.3	35.5	82.1	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.7	9.2	DB	1.4	98.0	8.6	4.2	-4.8	101.0	-9.9	103.4	-14.7	105.8	-21.0	108.8	(4)
(5)			WB	1.2	78.4	8.7	3.2	-5.0	80.7	-10.1	82.6	-15.0	84.4	-21.2	86.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	31.9	36.6	44.7	52.9	61.3	68.0	70.4	70.5	61.6	54.5	44.5	34.5		
	DBStd	15.38	8.75	8.56	9.02	7.43	6.43	6.46	5.30	5.73	5.84	7.73	8.57	8.00		
	HDD50	1948	562	379	212	55	3	1	0	0	1	44	208	483		
	HDD65	5067	1025	796	631	364	149	45	14	18	132	332	617	944		
	CDD50	2934	2	3	46	143	353	540	632	636	350	184	41	4		
	CDD65	578	0	0	0	2	34	135	180	189	31	7	0	0		
	CDH74	6892	0	0	15	116	575	1531	2000	2090	438	127	0	0		
	CDH80	2696	0	0	2	16	156	595	865	944	102	16	0	0		
(14) Wind	WSAvg	3.2	2.9	3.3	3.9	4.0	3.5	3.2	3.1	2.8	2.8	2.6	3.1	3.0		
(15) Precipitation	PrecAvg	35.6	2.4	2.2	2.0	3.0	3.6	4.1	4.1	3.1	2.8	2.1	3.0	3.1		
	PrecMax	48.5	5.2	4.7	5.3	5.4	8.8	7.0	11.4	8.4	5.9	5.4	5.5	6.2		
	PrecMin	23.6	0.4	0.5	0.3	0.7	0.9	1.8	0.6	0.6	0.8	0.0	0.5	0.2		
	PrecStd	4.9	1.3	1.2	1.3	1.1	2.0	1.6	2.3	1.9	1.2	1.4	1.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	66.1	75.5	82.3	87.9	93.3	96.9	100.2	87.5	82.3	71.4	62.9		
		MCWB	51.6	50.9	55.4	61.1	68.2	70.0	69.0	69.6	67.8	64.8	58.3	52.1		
	2%	DB	55.2	60.9	70.2	77.1	84.3	89.8	92.5	93.3	82.7	77.3	67.6	56.7		
		MCWB	47.7	48.7	53.3	59.1	66.2	70.0	70.5	70.4	66.2	61.9	55.6	49.0		
	5%	DB	48.5	55.6	66.3	73.2	80.7	87.4	88.9	89.5	79.1	73.5	64.0	51.4		
		MCWB	44.2	46.5	52.2	56.6	64.3	70.2	70.5	70.6	65.0	61.2	54.1	46.4		
	10%	DB	43.1	50.1	62.3	68.3	76.6	83.6	84.6	85.7	75.3	68.7	59.3	46.5		
		MCWB	41.1	43.7	51.0	55.2	63.1	68.5	69.2	69.7	63.4	59.2	52.2	43.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	52.7	60.6	65.0	72.7	76.3	77.2	77.7	72.7	67.0	59.8	53.0		
		MCDB	60.1	61.5	67.9	75.4	81.4	86.4	85.8	88.2	80.6	75.7	67.3	60.2		
	2%	WB	48.5	49.7	55.5	61.4	68.3	73.5	74.5	74.9	68.8	64.3	57.2	50.0		
		MCDB	53.9	59.4	65.8	72.0	79.1	83.9	85.4	86.9	78.6	74.1	64.6	55.4		
	5%	WB	44.8	47.3	53.6	59.2	66.0	71.3	72.2	72.8	66.5	62.3	54.6	46.9		
		MCDB	47.9	54.7	63.5	69.2	76.8	82.1	83.7	83.9	74.9	70.9	62.4	50.8		
	10%	WB	41.1	44.4	51.4	57.0	63.7	69.6	70.2	70.8	64.5	60.2	52.2	43.9		
		MCDB	43.5	49.5	61.2	65.6	73.4	81.1	81.7	80.6	72.7	67.1	59.5	46.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	18.4	22.7	23.0	23.6	23.3	24.0	25.1	22.6	23.7	18.1	14.6		
		MCDBR	25.0	29.5	31.8	32.7	32.2	30.4	32.0	34.0	30.9	31.5	27.2	24.2		
		MCWBR	17.7	18.9	17.0	16.7	15.9	13.7	13.3	14.4	15.7	17.2	16.5	17.4		
	5% WB	MCDBR	23.2	27.1	27.4	27.5	28.6	27.3	27.7	27.4	26.9	27.8	24.1	21.4		
		MCWBR	17.1	18.1	16.2	15.9	16.0	13.9	14.4	14.7	14.9	15.8	15.3	16.2		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.361	0.397	0.453	0.444	0.440	0.442	0.442	0.424	0.388	0.349	0.330		
	taud		2.375	2.271	2.221	2.084	2.159	2.219	2.214	2.212	2.236	2.332	2.396	2.385		
	Ebn at Noon		246	257	263	257	263	265	262	257	251	245	237	233		
	Edh at Noon		25	33	39	49	47	44	44	42	38	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		417	641	1021	1361	1680	1869	1929	1729	1193	833	503	343		
	RadStd		62	86	129	162	158	172	154	186	139	98	49	57		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+1.19	N/A	N/A	N/A	+0.67	N/A	-211	-331	+209	N/A

Nomenclature: See separate page